



CSA0501-DATABASE MANAGEMENT SYSTEMS

LAB EXPERIMENTS

DATE: 20/07/2026

EXPERIMENT 7 -- SELECT with various clause – GROUP BY, HAVING, ORDER BY

AIM:

To retrieve and organize records from the **student_cgpa** table using the **GROUP BY**, **HAVING**, and **ORDER BY** clauses in MySQL Workbench.

PROCEDURE:

1. Open MySQL Workbench and connect to the MySQL server.
2. Select the required database using the USE command.
3. Create the student_cgpa table and insert the required records.
4. Display the table contents using the SELECT statement.
5. Use the **GROUP BY** clause to group records based on the department.
6. Apply the **HAVING** clause to filter grouped results.
7. Use the **ORDER BY** clause to sort records in ascending and descending order.
8. Execute each query and verify the output in the Result Grid.
9. Observe the grouped, filtered, and sorted results.
10. Conclude the experiment after successful execution.

PROGRAM:

1. CREATE TABLE

```
CREATE TABLE student_cgpa (  
    reg_no VARCHAR(15) PRIMARY KEY,  
    student_name VARCHAR(50) NOT NULL,  
    department VARCHAR(20),  
    no_of_subs_pass INT,  
    no_of_subs_fail INT,  
    cgpa DECIMAL(3,2)  
);
```

Then Insert Values:

```
INSERT INTO student_cgpa  
(reg_no, student_name, department, no_of_subs_pass, no_of_subs_fail, cgpa)
```

VALUES

```
('192411123','Deepthi','CSE',22,2,8.71),  
('192512121','Varun','IT',27,1,9.23),  
('192312117','Aadhya','ECE',25,3,8.97),  
('192411227','Dimple','CSE',31,0,9.37),  
('192820896','Mohan','EEE',18,9,6.48),  
('192411122','Yogitha','CSE',31,1,9.01),  
( '192516871','Sadhya','IT',25,5,7.22),  
('192111101','Rahul','MECH',30,12,5.14),  
('192411010','Alekhyia','CSE',35,2,9.01),  
('192511135','Rahul','CIVIL',37,0,9.72);
```

```
SELECT * FROM student_cgpa;
```

OUTPUT:

reg_no	student_name	department	no_of_subs_pass	no_of_subs_fail	cgpa
192111101	Rahul	MECH	30	12	5.14
192312117	Aadhya	ECE	25	3	8.97
192411010	Alekhyia	CSE	35	2	9.01
192411122	Yogitha	CSE	31	1	9.01
192411123	Deepthi	CSE	22	2	8.71
192411227	Dimple	CSE	31	0	9.37
192511135	Rahul	CIVIL	37	0	9.72
192512121	Varun	IT	27	1	9.23
192516871	Sadhya	IT	25	5	7.22
192820896	Mohan	EEE	18	9	6.48

#	Time	Action	Message	Duration / Fetch
1	22:41:47	CREATE TABLE student_cgpa (reg_no VARCHAR(15) PRIMARY KEY, student_name VARCHAR(50) NO...	0 row(s) affected	0.031 sec
2	22:44:31	INSERT INTO student_cgpa (reg_no, student_name, department, no_of_subs_pass, no_of_subs_fail, cgpa) VA...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
3	22:46:18	SELECT * FROM student_cgpa LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec

GROUP BY

1. Count the number of students in each department.

```
SELECT department, COUNT(*) AS Number_of_Students  
FROM student_cgpa  
GROUP BY department;
```

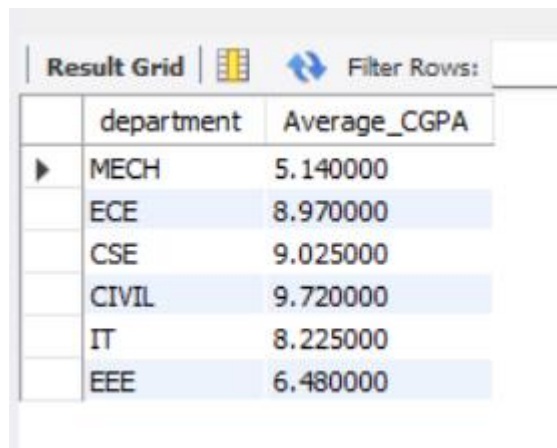
OUTPUT:

department	Number_of_Students
MECH	1
ECE	1
CSE	4
CIVIL	1
IT	2
EEE	1

2. Find the average CGPA of each department.

```
SELECT department, AVG(cgpa) AS Average_CGPA
FROM student_cgpa
GROUP BY department;
```

OUTPUT:



The screenshot shows a 'Result Grid' window with a table containing two columns: 'department' and 'Average_CGPA'. The table lists six departments with their corresponding average CGPA values: MECH (5.140000), ECE (8.970000), CSE (9.025000), CIVIL (9.720000), IT (8.225000), and EEE (6.480000). The window also includes a 'Filter Rows' button and a small icon.

department	Average_CGPA
MECH	5.140000
ECE	8.970000
CSE	9.025000
CIVIL	9.720000
IT	8.225000
EEE	6.480000

3. Find the maximum CGPA in each department.

```
SELECT department, MAX(cgpa) AS Maximum_CGPA
FROM student_cgpa
GROUP BY department;
```

OUTPUT:



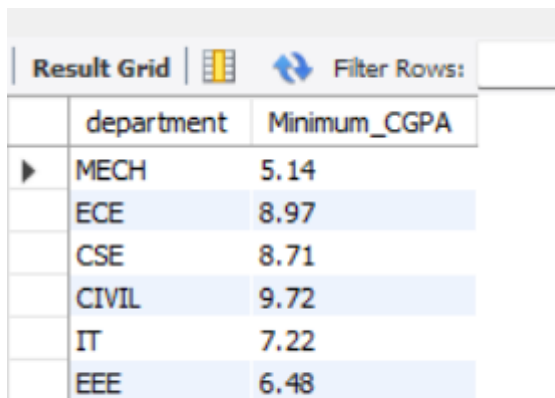
The screenshot shows a 'Result Grid' window with a table containing two columns: 'department' and 'Maximum_CGPA'. The table lists six departments with their corresponding maximum CGPA values: MECH (5.14), ECE (8.97), CSE (9.37), CIVIL (9.72), IT (9.23), and EEE (6.48). The window also includes a 'Filter Rows' button and a small icon.

department	Maximum_CGPA
MECH	5.14
ECE	8.97
CSE	9.37
CIVIL	9.72
IT	9.23
EEE	6.48

4. Find the minimum CGPA in each department.

```
SELECT department, MIN(cgpa) AS Minimum_CGPA
FROM student_cgpa
GROUP BY department;
```

OUTPUT:



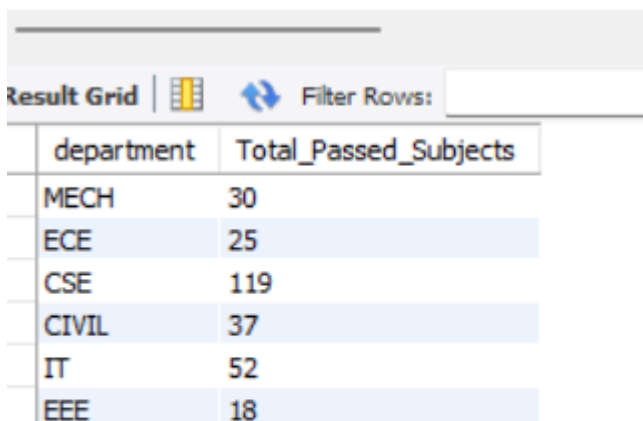
The screenshot shows a 'Result Grid' with a 'Filter Rows' button. The table has two columns: 'department' and 'Minimum_CGPA'. The data is as follows:

department	Minimum_CGPA
MECH	5.14
ECE	8.97
CSE	8.71
CIVIL	9.72
IT	7.22
EEE	6.48

5. Find the total number of passed subjects in each department.

```
SELECT department, SUM(no_of_subs_pass) AS Total_Passed_Subjects
FROM student_cgpa
GROUP BY department;
```

OUTPUT:



The screenshot shows a 'Result Grid' with a 'Filter Rows' button. The table has two columns: 'department' and 'Total_Passed_Subjects'. The data is as follows:

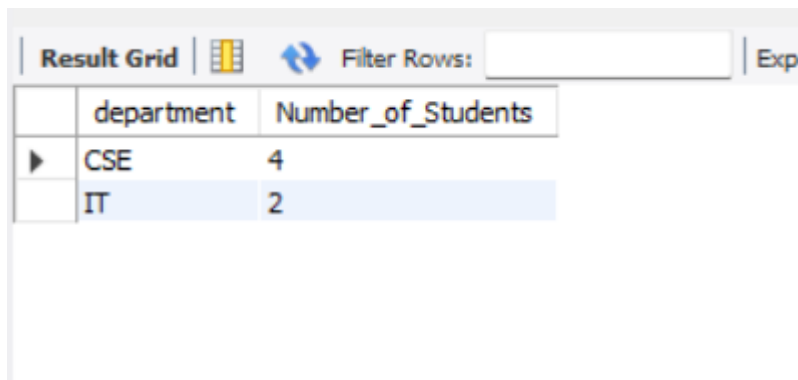
department	Total_Passed_Subjects
MECH	30
ECE	25
CSE	119
CIVIL	37
IT	52
EEE	18

HAVING

6. Display departments having more than one student.

```
SELECT department, COUNT(*) AS Number_of_Students
FROM student_cgpa
GROUP BY department
HAVING COUNT(*) > 1;
```

OUTPUT:



The screenshot shows a database result grid with a toolbar at the top containing 'Result Grid', a grid icon, a 'Filter Rows' button with a double arrow icon, a text input field, and an 'Export' button. The table has two columns: 'department' and 'Number_of_Students'. The first row is 'CSE' with 4 students, and the second row is 'IT' with 2 students.

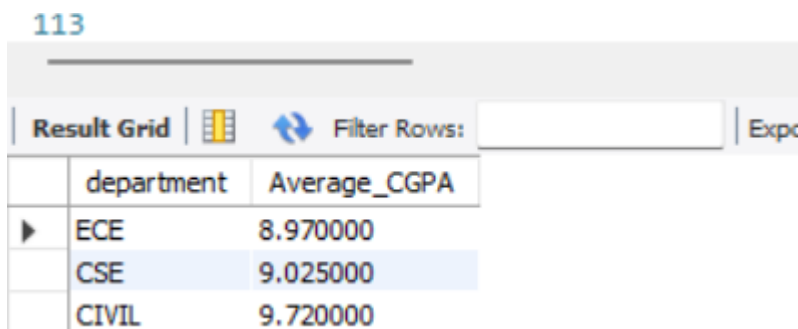
	department	Number_of_Students
▶	CSE	4
	IT	2

7. Display departments whose average CGPA is greater than 8.50.

```
SELECT department, AVG(cgpa) AS Average_CGPA
FROM student_cgpa
GROUP BY department
HAVING AVG(cgpa) > 8.50;
```

OUTPUT:

113



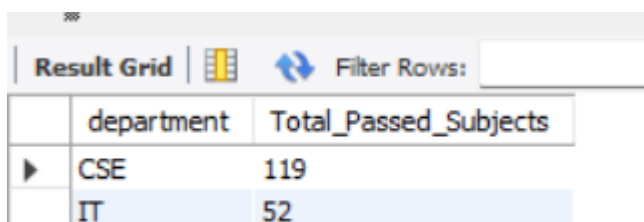
The screenshot shows a database result grid with a toolbar at the top containing 'Result Grid', a grid icon, a 'Filter Rows' button with a double arrow icon, a text input field, and an 'Export' button. The table has two columns: 'department' and 'Average_CGPA'. The first row is 'ECE' with an average CGPA of 8.970000, the second row is 'CSE' with 9.025000, and the third row is 'CIVIL' with 9.720000.

	department	Average_CGPA
▶	ECE	8.970000
	CSE	9.025000
	CIVIL	9.720000

8. Display departments whose total passed subjects are greater than 50.

```
SELECT department, SUM(no_of_subs_pass) AS Total_Passed_Subjects
FROM student_cgpa
GROUP BY department
HAVING SUM(no_of_subs_pass) > 50;
```

OUTPUT:



The screenshot shows a database result grid with a toolbar at the top containing 'Result Grid', a grid icon, a 'Filter Rows' button with a double arrow icon, a text input field, and an 'Export' button. The table has two columns: 'department' and 'Total_Passed_Subjects'. The first row is 'CSE' with 119 passed subjects, and the second row is 'IT' with 52 passed subjects.

	department	Total_Passed_Subjects
▶	CSE	119
	IT	52

ORDER BY

9. Display all students in ascending order of Registration Number.

```
SELECT *  
FROM student_cgpa  
ORDER BY reg_no ASC;
```

OUTPUT:

Result Grid

Filter Rows:

Edit:

Export/Import:

	reg_no	student_name	department	no_of_subs_pass	no_of_subs_fail	cgpa
▶	192111101	Rahul	MECH	30	12	5.14
	192312117	Aadhya	ECE	25	3	8.97
	192411010	Alekhyia	CSE	35	2	9.01
	192411122	Yogitha	CSE	31	1	9.01
	192411123	Deepthi	CSE	22	2	8.71
	192411227	Dimple	CSE	31	0	9.37
	192511135	Rahul	CIVIL	37	0	9.72
	192512121	Varun	IT	27	1	9.23
	192516871	Sadhya	IT	25	5	7.22
	192820896	Mohan	EEE	18	9	6.48

10. Display all students in descending order of CGPA.

```
SELECT *  
FROM student_cgpa  
ORDER BY cgpa DESC;
```

OUTPUT:

Result Grid



Filter Rows:

Edit:







Export/Import:



	reg_no	student_name	department	no_of_subs_pass	no_of_subs_fail	cgpa
▶	192511135	Rahul	CIVIL	37	0	9.72
	192411227	Dimple	CSE	31	0	9.37
	192512121	Varun	IT	27	1	9.23
	192411010	Alekhyia	CSE	35	2	9.01
	192411122	Yogitha	CSE	31	1	9.01
	192312117	Aadhya	ECE	25	3	8.97
	192411123	Deepthi	CSE	22	2	8.71
	192516871	Sadhya	IT	25	5	7.22
	192820896	Mohan	EEE	18	9	6.48
	192111101	Rahul	MECH	30	12	5.14

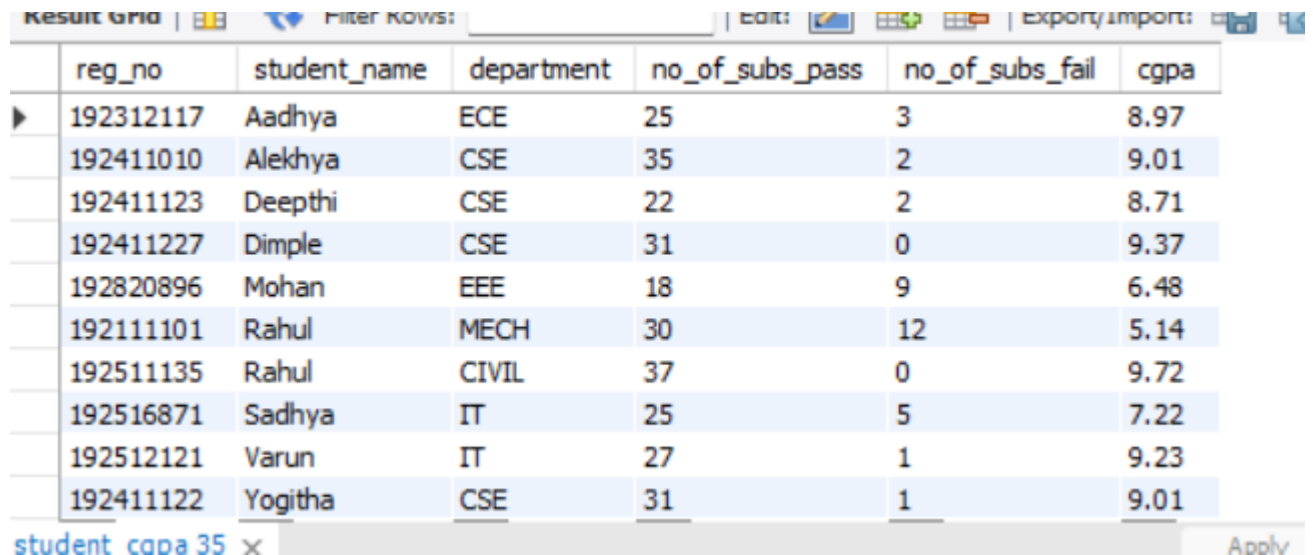
student_cgpa 34 x

Apply

11. Display students in ascending order of Student Name.

```
SELECT *  
FROM student_cgpa  
ORDER BY student_name ASC;
```

OUTPUT:



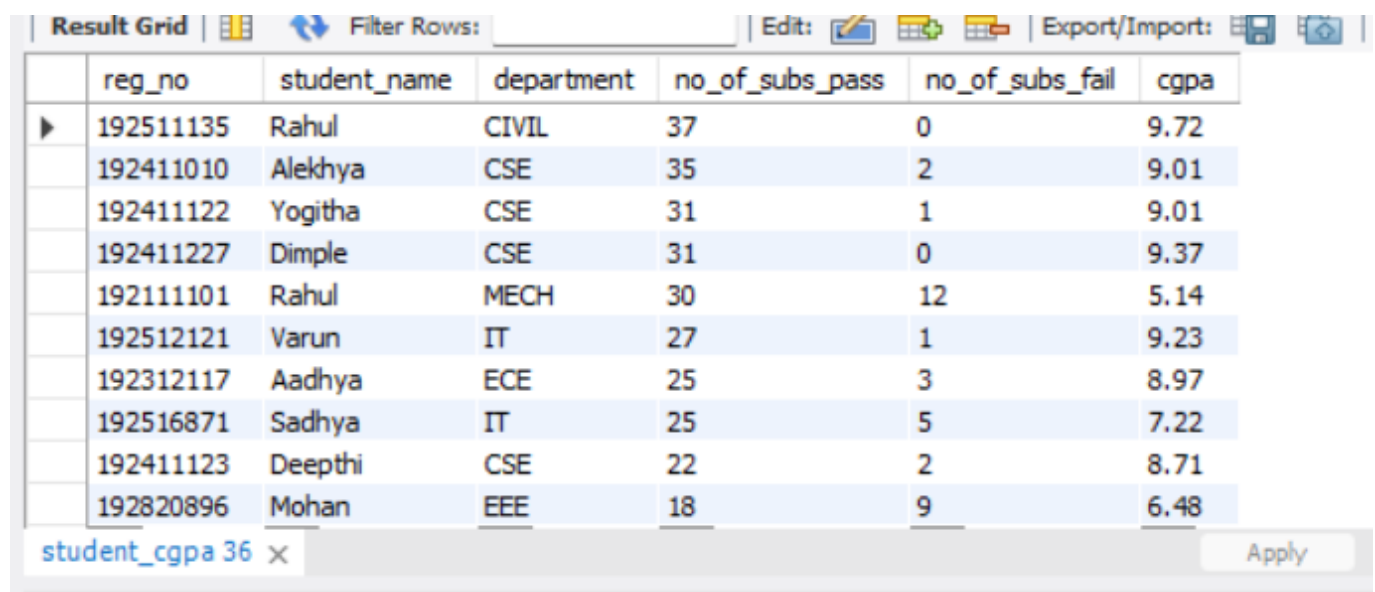
The screenshot shows a 'Result Grid' window with a table of student data. The table has columns: reg_no, student_name, department, no_of_subs_pass, no_of_subs_fail, and cgpa. The data is sorted by student_name in ascending order. The status bar at the bottom indicates 'student_cgpa 35' and an 'Apply' button.

reg_no	student_name	department	no_of_subs_pass	no_of_subs_fail	cgpa
192312117	Aadhya	ECE	25	3	8.97
192411010	Alekhya	CSE	35	2	9.01
192411123	Deepthi	CSE	22	2	8.71
192411227	Dimple	CSE	31	0	9.37
192820896	Mohan	EEE	18	9	6.48
192111101	Rahul	MECH	30	12	5.14
192511135	Rahul	CIVIL	37	0	9.72
192516871	Sadhya	IT	25	5	7.22
192512121	Varun	IT	27	1	9.23
192411122	Yogitha	CSE	31	1	9.01

12. Display students in descending order of Number of Passed Subjects.

```
SELECT *  
FROM student_cgpa  
ORDER BY no_of_subs_pass DESC;
```

OUTPUT:



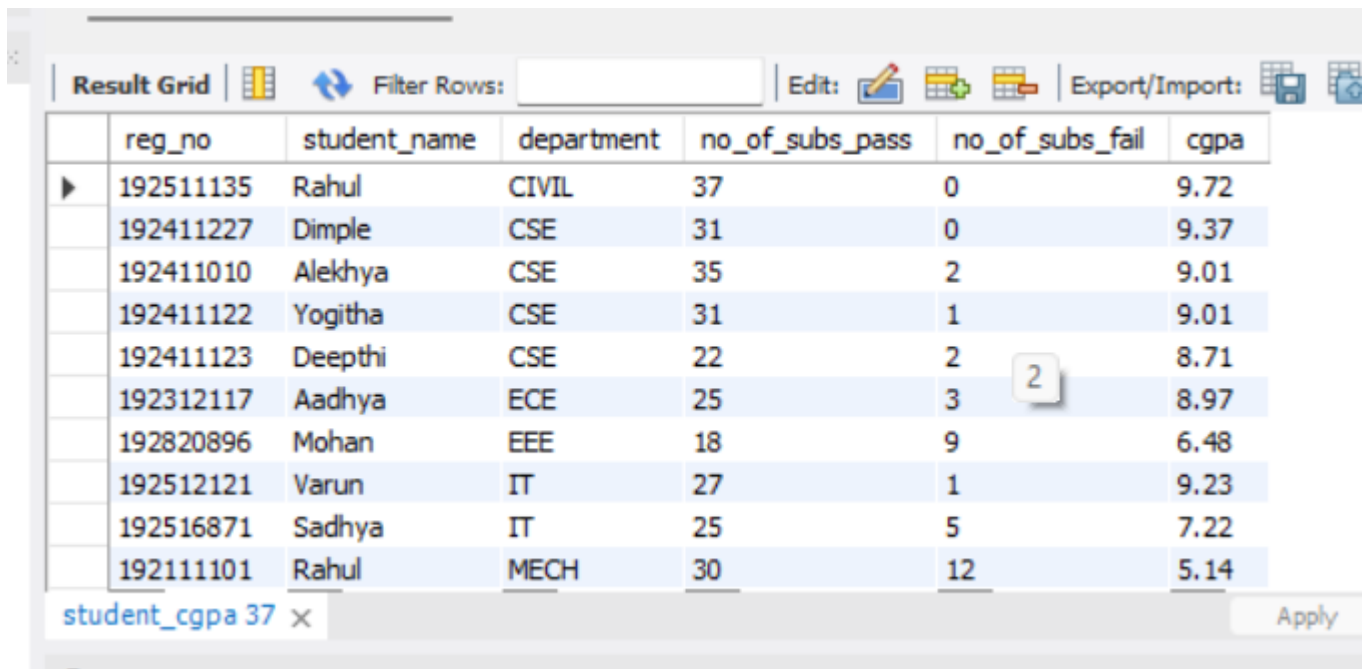
The screenshot shows a 'Result Grid' window with a table of student data. The table has columns: reg_no, student_name, department, no_of_subs_pass, no_of_subs_fail, and cgpa. The data is sorted by no_of_subs_pass in descending order. The status bar at the bottom indicates 'student_cgpa 36' and an 'Apply' button.

reg_no	student_name	department	no_of_subs_pass	no_of_subs_fail	cgpa
192511135	Rahul	CIVIL	37	0	9.72
192411010	Alekhya	CSE	35	2	9.01
192411122	Yogitha	CSE	31	1	9.01
192411227	Dimple	CSE	31	0	9.37
192111101	Rahul	MECH	30	12	5.14
192512121	Varun	IT	27	1	9.23
192312117	Aadhya	ECE	25	3	8.97
192516871	Sadhya	IT	25	5	7.22
192411123	Deepthi	CSE	22	2	8.71
192820896	Mohan	EEE	18	9	6.48

13. Display students ordered first by Department and then by CGPA in descending order.

```
SELECT *  
FROM student_cgpa  
ORDER BY department ASC, cgpa DESC;
```

OUTPUT:



The screenshot shows the MySQL Workbench interface with the 'Result Grid' tab active. The grid displays 11 rows of student data, ordered by department (ASC) and then by CGPA (DESC). The columns are: reg_no, student_name, department, no_of_subs_pass, no_of_subs_fail, and cgpa. The data is as follows:

reg_no	student_name	department	no_of_subs_pass	no_of_subs_fail	cgpa
192511135	Rahul	CIVIL	37	0	9.72
192411227	Dimple	CSE	31	0	9.37
192411010	Alekhyia	CSE	35	2	9.01
192411122	Yogitha	CSE	31	1	9.01
192411123	Deepthi	CSE	22	2	8.71
192312117	Aadhya	ECE	25	3	8.97
192820896	Mohan	EEE	18	9	6.48
192512121	Varun	IT	27	1	9.23
192516871	Sadhya	IT	25	5	7.22
192111101	Rahul	MECH	30	12	5.14

At the bottom of the window, the text 'student_cgpa 37' is visible next to a close button (X), and an 'Apply' button is on the right.

RESULT

Thus, the records from the student_cgpa table were successfully retrieved and organized using the GROUP BY, HAVING, and ORDER BY clauses in MySQL Workbench.